PERFECT COMPLEXES ARE EXACTLY THE PSEUDO-COHERENT COMPLEXES OF FINITE TOR DIMENSION, WITH A BOUNDED PROJECTIVE MODEL

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Editorial note. The counted statement span has two enumerated conditions followed by a further assertion, all delivered verbatim. The counted proof span uses the exact name E^bullet introduced in the statement and chooses E^bullet = tau_{>=a}F^bullet; that choice is delivered verbatim. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Pseudo-coherence and finite tor dimension of a perfect complex, and the tor amplitude statement. invoked at proof lines 18902-18903. Gives the implication (1) => (2).
- Existence of a bounded-above complex of finite free modules quasi-isomorphic to a pseudo-coherent complex. invoked at proof lines 18906-18908. Produces the model F^bullet whose truncation is the required E^bullet.
- A finitely presented flat module is finite projective, and the last-term-flat criterion. invoked at proof lines 18911-18913. Shows the single non-free term E^a is finite projective.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 16883: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16883–16883. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16048: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16048–16048. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 16906: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16906–16906. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $K^\bullet$ be an object of $D(R)$. The following are equivalent*

(1) *$K^\bullet$ is perfect, and*
(2) *$K^\bullet$ is pseudo-coherent and has finite tor dimension.*

If (1) and (2) hold and $K^\bullet$ has tor-amplitude in $[a, b]$, then $K^\bullet$ is quasi-isomorphic to a complex $E^\bullet$ of finite projective R-modules with $E^i = 0$ for $i \notin [a, b]$.

Proof. It is clear that (1) implies (2), see Lemmas 16048 and 16906. Assume (2) holds and that $K^\bullet$ has tor-amplitude in $[a, b]$. In particular, $H^i(K^\bullet) = 0$ for $i > b$. Choose a complex $F^\bullet$ of finite free R-modules with $F^i = 0$ for $i > b$ and a quasi-isomorphism $F^\bullet \to K^\bullet$ (Lemma 16048). Set $E^\bullet = \tau_{\geq a} F^\bullet$. Note that E^i is finite free except E^a which is a finitely presented R-module. By Lemma 16883 E^a is flat. Hence by Algebra, Lemma 00NX we see that E^a is finite projective. □